

A Region-of-Interest (ROI) Driven Deep Learning Pipeline for Precise and Explainable Brain Tumor Classification

1. Introduction

Brain tumors are among the most aggressive diseases, where early and accurate diagnosis is critical for patient survival. Magnetic Resonance Imaging (MRI) is the standard modality for diagnosis. However, manual interpretation by radiologists is time-consuming, subjective, and prone to human error. While Artificial Intelligence (AI) has shown promise in automating this process, most existing systems function as "Black Boxes"—they provide a prediction without explaining *why*. Furthermore, many existing models process the entire MRI scan, often leading to false positives caused by healthy features like the skull, eyes, or background noise.

2. Literature Review & Existing Gaps

We have analyzed recent research papers in this domain, including [Tiwari et al. \(2022\)](#) and [Chattopadhyay et al. \(2022\)](#). While these studies achieve high accuracy using Convolutional Neural Networks (CNNs), we identified specific limitations (Research Gaps) that our project aims to address:

The Identified Gaps:

- 1. The "Whole Image" Bias:** Most existing models feed the **entire MRI slice** into the classifier. This confuses the AI, as it may learn to classify based on the shape of the skull or the background darkness rather than the actual tumor texture.
- 2. Lack of Explainability (The "Black Box" Problem):** Existing papers focus solely on accuracy (e.g., "99% accuracy"). They do not provide visual evidence to the doctor proving *where* the model is looking. In a clinical setting, a doctor cannot trust a label without visual verification.
- 3. Ambiguity in Segmentation:** Many papers claim "detection" but essentially perform simple binary classification (Tumor vs. No Tumor) without actually delineating the tumor boundaries.

3. Problem Statement

The core problem we are solving is the **lack of precision and trust** in standard medical AI models. A standard CNN might classify an image correctly, but for the wrong reasons (e.g., detecting a skull artifact). There is a need for a pipeline that first **isolates** the pathology (tumor) from the healthy brain and then **explains** its decision to the radiologist.

4. Proposed Methodology (The Solution)

We propose a novel **Multi-Stage ROI Pipeline** that mimics the workflow of a human radiologist. Instead of a single model, we will implement a dual-model architecture enhanced with Explainable AI (XAI).

The Workflow:

- **Step 1: Tumor Segmentation (The "Eye"):**
 - **Model:** U-Net (or Attention U-Net).
 - **Function:** This model scans the full MRI and generates a binary **Mask**, identifying the exact pixel location of the tumor.
- **Step 2: ROI Cropping (The "Focus"):**
 - **Logic:** Using Computer Vision (OpenCV), we will use the mask from Step 1 to mathematically crop the original MRI.
 - **Result:** A clean image containing **only the tumor**, with the skull and background removed.
- **Step 3: Fine-Grained Classification (The "Brain"):**
 - **Model:** ResNet50 (Transfer Learning).
 - **Function:** The cropped image is fed into this classifier to predict the specific tumor type: **Meningioma, Glioma, or Pituitary Tumor**.
- **Step 4: Explainability (The "Proof"):**
 - **Technique:** Grad-CAM (Gradient-weighted Class Activation Mapping).
 - **Output:** We will generate a heatmap overlay on the tumor. This visualizes the specific regions the model used to make its decision, validating that the AI is looking at the tumor cells and not the background.

5. Novelty & Contribution

Our project moves beyond the basic "Hello World" of medical imaging by implementing a clinically viable pipeline. Our specific contributions are:

1. **Noise Reduction via ROI:** By physically removing the skull and background before classification, we eliminate the primary source of false positives found in previous research.
2. **Trust via XAI:** We are integrating **Grad-CAM** to provide visual "proof of correctness," bridging the gap between computer science and medical practitioners.
3. **Hybrid Architecture:** Combining a Segmenter (U-Net) and a Classifier (ResNet50) into a single cohesive pipeline is a significant step up from standard single-model projects.

6. Tools & Technologies

- **Language:** Python
- **Deep Learning Frameworks:** TensorFlow / Keras
- **Image Processing:** OpenCV, NumPy

- **Dataset:** Kaggle Brain Tumor MRI Dataset (Containing Images and Masks)
- **Visualization:** Matplotlib (for Heatmaps)

7. Expected Outcomes

- A working prototype that takes a raw MRI scan as input.
- **Output 1:** A segmented mask showing the tumor boundary.
- **Output 2:** The specific classification label (e.g., "Glioma").
- **Output 3:** An XAI Heatmap validating the decision.
- A comparative analysis showing that our **ROI-based approach** outperforms standard **Whole-Image approaches** in terms of reliability.